Comparing Safety-Removal Subspaces in Aligned LLMs

Midterm Project Update

Group 6: Evan Scamehorn, Kyle Sha, Adam Venton, Zeke Mackay, and Calvin Kosmatka

https://github.com/evan203/nlp-project-proposal

Research question:

- Do different linear safety-removal methods recover the same refusal mechanism?
- When a method removes refusal behavior, how much utility is preserved?
- Do Geometry-style RepInd profiles show independent refusal directions?

Baseline Method:

- Difference-in-means (DIM) [1]
  - Mean response activation difference between harmful and harmless prompts
- Refusal Cone Optimization (RCO) [2]
  - Use gradient descent to generate multiple basis vectors representing safety mechanisms
- ActSVD safety/utility ranks [3]
  - Perform Singular Value Decomposition on model weights to identify safety/utility-critical low-rank matrices
- Mode Subspace Overlap (MSO) [4]
  - Performs SVD to quantify overlap between subspaces
- Representational Independence (RepInd) [2]
  - Tests whether ablating one direction changes another direction's layerwise cosine profile

Experiment setup:

- Target model: Llama-3.1-8B-Instruct
- Safety eval: JailbreakBench attack success rate
- Utility eval: harmless Alpaca compliance + Pile/Alpaca perplexity
- Geometry eval:
  - MSO between DIM refusal direction and ActSVD weight-delta subspaces [4]
  - MSO between DIM safety directions and harmless-instruction utility PCA subspaces
  - RepInd profile changes before/after direction ablation [2]

Findings:

- DIM ablation raises JBB ASR from 0.16 to 1.00 with little perplexity change.
- ActSVD raises JBB ASR to 0.63 but causes larger Pile/Alpaca perplexity degradation.
- DIM vs ActSVD MSO is near random for most layers, with a mild hotspot around layer 10.
- Direct safety-vs-utility overlap is above random: rank-8 mean MSO = 0.192 vs 0.00195 random baseline.
- RepInd profile test is asymmetric: ablating DIM strongly changes one derived basis profile, but ablating that basis barely changes DIM.

Cones/RepInd Next Step:

- Current RepInd run uses DIM-derived cone-basis candidates.
- Full optimized cone claim requires running `scripts/run_rco.sh`.
- Then compare DIM, RDO, orthogonal-RDO, RepInd, and cone basis directions with the same RepInd script.
- Evaluate cone samples against DIM and ActSVD on the same safety/utility benchmark.

Contribution:

- Describe each team member’s contribution.
- Provide the percentage of contributions within the group. For example, Member 1 contributes 40% of efforts. Member 2 20%, Member 3 20%, Member 4 20%.

References

[1] A. Arditi *et al.*, "Refusal in Language Models Is Mediated by a Single Direction," in *Advances in Neural Information Processing Systems*, A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, Eds., Curran Associates, Inc., 2024, pp. 136037–136083. doi: 10.52202/079017-4322.

[2] T. Wollschläger, J. Elstner, S. Geisler, V. Cohen-Addad, S. Günnemann, and J. Gasteiger, "The Geometry of Refusal in Large Language Models: Concept Cones and Representational Independence," in *Proceedings of the 42nd International Conference on Machine Learning*, A. Singh, M. Fazel, D. Hsu, S. Lacoste-Julien, F. Berkenkamp, T. Maharaj, K. Wagstaff, and J. Zhu, Eds., in Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 66945–66970. [Online]. Available: https://proceedings.mlr.press/v267/wollschlager25a.html

[3] B. Wei *et al.*, "Assessing the Brittleness of Safety Alignment via Pruning and Low-Rank Modifications," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, in Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 52588–52610. [Online]. Available: https://proceedings.mlr.press/v235/wei24f.html

[4] K. Ponkshe, S. Shah, R. Singhal, and P. Vepakomma, "Safety Subspaces are Not Linearly Distinct: A Fine-Tuning Case Study," in *Proceedings of the Fourteenth International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://arxiv.org/abs/2505.14185

[5] R. Taori *et al.*, "Stanford Alpaca: An Instruction-following LLaMA model." GitHub, 2023.

[6] J. Ji *et al.*, "BeaverTails: Towards Improved Safety Alignment of LLM via a Human-Preference Dataset." [Online]. Available: https://arxiv.org/abs/2307.04657

[7] P. Chao *et al.*, "JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models," in *NeurIPS Datasets and Benchmarks Track*, 2024. [Online]. Available: https://arxiv.org/abs/2404.01318

[8] L. Gao *et al.*, "The Pile: An 800GB Dataset of Diverse Text for Language Modeling," *arXiv preprint arXiv:2101.00027*, 2020, [Online]. Available: https://arxiv.org/abs/2101.00027

[9] M. Mazeika *et al.*, "HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. [Online]. Available: https://arxiv.org/abs/2402.04249

[10] L. Jiang *et al.*, "WildTeaming at Scale: From In-the-Wild Jailbreaks to (Adversarially) Safer Language Models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [Online]. Available: https://arxiv.org/abs/2406.18510